Factorising Out Terms

Exam code: 0607

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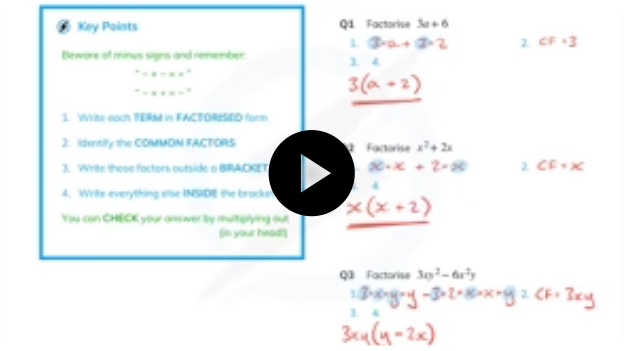

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Basic factorising

What is factorisation?

- A **factorised** expression is one written as the **product** (**multiplication**) of two, or more, terms (**factors**)
  - $3(x + 2)$ is factorised
    - It is $3 \times (x + 2)$
  - $3x + 6$ is **not** factorised
  - $3xy$ is factorised
    - It is $3 \times x \times y$
  - Numbers can also be factorised
    - $12 = 2 \times 2 \times 3$
- In algebra, factorisation is the **reverse** of **expanding brackets**
  - It's putting it into brackets, rather than removing brackets

How do I factorise two terms?

- To factorise $12x^2 + 18x$
  - Find the highest common factor of the **number** parts
    - 6
  - Find the highest common factor of the **algebra** parts
    - x

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- $12x^2 + 18x$ is the same as $6x \times 2x + 6x \times 3$
  - Using the highest common factor
- **Take out** the highest common factor
  - Write it **outside** a set of **brackets**
  - Put the remaining terms, $2x + 3$, inside the brackets
- This gives the answer
  - $6x(2x + 3)$

- To factorise an expression containing multiple variables, e.g. $2a^3b - 4a^2b^2$
  - Use the same approach as above
  - Find the highest common factor of the **number** parts
    - 2
  - Find the highest common factor of the **algebra** parts
    - a and b appear in both terms
    - The highest common factor of a^3 and a^2 is a^2
    - The highest common factor of b and b^2 is b
    - a^2b
  - Multiply both to get the overall **highest common factor**
    - $2a^2b$
  - $2a^3b - 4a^2b^2$ is the same as $2a^2b \times a - 2a^2b \times 2b$
    - Using the highest common factor
  - **Take out** the highest common factor
    - Write it **outside** a set of **brackets**
    - Put the remaining terms, $a - 2b$, inside the brackets
  - This gives the answer
    - $2a^2b(a - 2b)$

Examiner Tips and Tricks

In the exam, check that your factorisation is correct by **expanding** the brackets.

Factorise means factorise **fully.**

$x(6x + 10)$ is not fully factorised but $2x(3x + 5)$ is.

Worked Example

(a) Factorise $5x + 15$
Answer:

Find the highest common factor of 5 and 15

$$5 \times x + 5 \times 3$$

Take out the 5 and put $x + 3$ in brackets

$$\mathbf{5(x + 3)}$$

(b) Factorise fully $30x^2 - 24x$

Answer:

Find the highest common factor of 30 and 24

$$6$$

Find the highest common factor of x^2 and x

$$x$$

Find the overall highest common factor by multiplying these together

$$6x$$

Think of each term as $6x \times$ something

$$6x \times 5x - 6x \times 4$$

Take out the $6x$ and put $5x - 4$ in brackets

$$\mathbf{6x(5x - 4)}$$

(c) Factorise fully $5x^4y + 25x^3y^3$

Answer:

Find the highest common factor of 5 and 25

$$5$$

Find the highest common factor of x^4 and x^3

$$x^3$$

Find the highest common factor of y and y^3

$$y$$

Find the overall highest common factor by multiplying these together

$$5x^3y$$

Think of each term as $5x^3y \times$ something

$$5x^3y \times x + 5x^3y \times 5y^2$$

Take out the $5x^3y$ and put $x + 5y^2$ in brackets

$$\mathbf{5x^3y(x + 5y^2)}$$

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